

Philip Bulsink

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SUMMARY

Senior analytical chemist with 10+ years of experience integrating advanced chromatography, spectroscopy, chemometrics, and scientific software development to solve complex characterization challenges in bioenergy and petroleum research. Recognized for leading international interlaboratory studies, developing novel analytical methods, and translating scientific findings into strategic advice for government and industry stakeholders.

CURRENT POSITION

Fuels Chemist (CH-03)

September 2014 — Present

Characterization Laboratory, CanmetENERGY-Ottawa, Natural Resources Canada

Ottawa, Ontario

- Perform detailed chemical analyses of complex sample matrices using advanced analytical techniques.
- Manage the full-lifecycle of a multi-vendor scientific instrumentation fleet, including acquisition, implementation, maintenance, calibration, troubleshooting, repair, and disposition.
- Pioneer analytical methods using advanced chemical techniques and custom-built chromatographic and spectroscopic equipment (such as GC-FID/MS, GCxGC-FID, GCxGC-FID/MS, FTIR, and Raman) to characterize biofuels and petroleum products, resolving complex analytical bottlenecks for research groups and industry partners.
- Develop custom chemometric algorithms and open-source R tools to automate complex spectroscopic data analysis and visualization across research projects.
- Translate complex chemometric data into executive summaries, high-level briefings, and actionable technical reports for non-technical stakeholders and interdisciplinary teams.
- Lead and co-author journal publications and presentations, contributing to the dissemination of research findings.
- Contribute to research funding proposals, strengthening experimental design and improving project outcomes.
- Represent the laboratory, department, and Canadian scientific interests at internal, national and international fora, including conferences, technical meetings, working groups, and standards boards.
- Champion data quality systems by serving as a CALA-certified ISO 17025 internal auditor, verifying complex analytical processes to ensure strict compliance and experimental reliability.
- Train and support junior staff and co-op students, fostering a culture of knowledge sharing and professional development.
- Act up as CH-04 with managerial and Sections 32 & 34 authority, as required.

Key Projects and Activities

Method Development Lead, SPME & Derivatization Analysis of Biorefinery Streams

2025 — Present

- Developed *in situ* derivatization and SPME workflows using Design of Experiment (DoE) principles to capture, stabilize, and quantify reactive carbonyl target compounds in solid and liquid biorefinery products.
- Prepared comprehensive technical reports detailing occupational exposure risks of different matrices.

Developer & Maintainer, PlotFTIR (<https://nrcan.github.io/PlotFTIR>)

2024 — Present

- Developed an R package for plotting and analyzing Fourier Transform Infrared (FTIR) and Raman spectroscopy data, enabling researchers to visualize and interpret complex spectral information effectively.

Author and Principal Investigator, International Round Robin Studies

2020 — 2025

- Lead international round-robin studies of bio-liquefaction oils on behalf of Canada at IEA Bioenergy Task34.
- Produced two first-author papers, fostered international collaboration and improved understanding of analytical methods.

Method Development Lead, GC-FID/MS Semiquantitative Analysis

2018 — 2022

- Built chromatographic method and custom data analysis algorithms to produce semiquantitative compositional reports with nearly 100% mass balance for complex samples.
- Validated method validity on petroleum, low carbon/renewable fungible fuels, and biofuel products.

Emergency Response and Building Evacuation Teams

2015 — Present

- Comprehensively trained in emergency medical and chemical response procedures, including advanced first aid, burn response, trauma/wound care, emergency triage, scene control and hazardous spill control operations.
- Perform backup dispatch, incident commander, and technical HAZMAT expert roles.
- Serve as Chief Building Evacuation Warden, responding to all facilities hazards.

PREVIOUS EXPERIENCE

Research Assistant (Co-Op)	2010 — 2011
DeNOx Group, CanmetENERGY-Ottawa, Natural Resources Canada	Ottawa, Ontario
- Investigated homogeneous catalysts for the reduction of NO_x in lean-burn diesel engine exhaust. - Scaled catalyst synthesis by three orders of magnitude. - Custom-built analytical instrumentation and analysis software to support research inquiries.	
Laboratory Teaching Assistant	September 2011 — May 2014
University of Ottawa / University of Waterloo	Ottawa / Waterloo, Ontario
Supervised undergraduate laboratories across analytical, organic, inorganic, and physical chemistry, instructing on advanced techniques and lab safety protocols.	

EDUCATION

University of Ottawa	2012 — 2014
<i>Master's of Science, Chemistry</i>	Ottawa, Ontario
- Thesis: "Rhenium (I) Terdentate Compounds: Theoretical and Experimental Investigations" - Seminar: "Recent Advancements in NO_x Abatement from Diesel Engine Emissions"	
University of Waterloo	2007 — 2012
<i>Bachelor's of Science, Honours Chemistry (Co-op), Music Minor</i>	Waterloo, Ontario
- Thesis: "Solid Sample Analysis by Microplasma Optical Emission Spectroscopy" - Achieved "Outstanding" co-op term ranking for each of the 5 work terms (2008-2012)	

EXTRACURRICULARS & COMMUNITY LEADERSHIP

Chair of Council (Executive Committee), Kanata Community Church	2024 — Present
Provide executive governance and administrative leadership for the church, managing council records, overseeing committee logistics, and supporting strategic operations and policy execution.	
Author & Maintainer, fldataR R Package	2023 — Present
Develop and maintain an open-source R package for Formula 1 data analytics, managing CRAN submission requirements, automated test coverage workflows, and community issue tracking on GitHub.	

AWARDS AND DISTINCTIONS

Scientific Innovation , Branch Award, CanmetENERGY-Ottawa	2026
Excellence in Science , Departmental Award, Natural Resources Canada	2023
Positive Workplace Impact , Sector Award, Energy Efficiency & Technology	2022
Innovation & Creativity , Branch Award, CanmetENERGY-Ottawa	2021
Dean's Scholarship , University of Ottawa	2015
Dean's Honour Roll , University of Waterloo; University of Ottawa	2011-2014
Recognition of Collaboration , Departmental Award, Natural Resources Canada	2012
Aileen Proudfoot Award , Branch Award, CanmetENERGY-Ottawa	2011

TECHNICAL SKILLS & EXPERTISE

- Analytical Instrumentation & Maintenance:**
 - Extensive hands-on operation, routine maintenance, hardware troubleshooting, and calibration of Agilent GC systems (FID, MS, FPD, PFPD, TCD), PAL autosamplers, and FTIR spectrometers.
- Methodologies & Synthesis:**
 - In situ* derivatization, SPME, sample preparation, synthetic chemistry techniques, Design of Experiments (DoE), chemometric modeling.
- Software & Data Analysis:**
 - R (package development, CRAN maintenance), Python, VBA, RDKit, Agilent MassHunter, Agilent ChemStation, Bruker TopSpin, Gaussian, TurboMole.
- Linguistic Profile:** English Native | French CBA

SELECTED PUBLICATIONS AND PRESENTATIONS

- **Bulsink, P.; et al.** (2026) “Characterization of Volatile Carbonyl Compounds in Biorefinery Streams using SPME-GC: Implications for Occupational Health and Safety” **Pyro2026 Conference**. Pisa, Italy
- Zhang, Y.; Monnier, J.; **Bulsink, P.; et al.** (2025). “Evaluation of diesel fuel production from bio-oils hydrodeoxygenation using unsupported MoS₂ catalysts” **Fuel Processing Technology**, 276, 108290. [10.1016/j.fuproc.2025.108290](https://doi.org/10.1016/j.fuproc.2025.108290).
- **Bulsink, P.; et al.** (2025). “Multidimensional Chromatography of Bio/Renewable Energy Products “ **Canadian Society of Chemistry Conference**. Ottawa, Ontario
- **Bulsink, P.; et al.** (2025). “Interlaboratory Study of Sample Homogeneity Impact on CHNS, Water, and ICP Analysis of Biomass Liquefaction Oils” **Energy & Fuels**, 39 (29), 14223-14236. [10.1021/acs.energyfuels.5c01309](https://doi.org/10.1021/acs.energyfuels.5c01309).
- **Bulsink, P.; et al.** (2023). “Quantification of components without direct calibration by GC-MS/PolyArc®-FID” **American Chemical Society Conference**. San Francisco, California
- **Bulsink, P.** (2021). “Gas Chromatographic and Mass Spectral Analysis in Characterization Lab” *CanmetENERGY-Ottawa Science Seminar*
- **Bulsink, P.; et al.** (2020). “Results of the IEA Bioenergy Round Robin on the Analysis of Heteroatoms in Biomass Liquefaction Oils” *CanmetENERGY-Ottawa Science Seminar*
- **Bulsink, P.; et al.** (2020). “Results of the International Energy Agency Bioenergy Round Robin on the Analysis of Heteroatoms in Biomass Liquefaction Oils” **Energy & Fuels**, 34 (9), 11123-11133. [10.1021/acs.energyfuels.0c02090](https://doi.org/10.1021/acs.energyfuels.0c02090).
- **Bulsink, P.; et al.** (2016). “Capturing Re(I) in a neutral N,N,N pincer Scaffold and resulting enhanced absorption of visible light” **Dalton Transactions** 45, 8885-8896. [10.1039/c6dt00661b](https://doi.org/10.1039/c6dt00661b).
- Stanciulescu, M.; **Bulsink, P.; et al.** (2014). “NH₃-TPD-MS study of Ce effect on the surface of Mn-or Fe-exchanged zeolites for selective catalytic reduction of NO_x by ammonia” **Applied Surface Science** 300, 201-207. [10.1016/j.apsusc.2014.01.175](https://doi.org/10.1016/j.apsusc.2014.01.175).
- Stanciulescu, M; Caravaggio, G.; Dobri, A.; Moir, J.; Burich, R.; Charland, J.-P.; **Bulsink, P.**; (2012). “Low-temperature selective catalytic reduction of NO_x with NH₃ over Mn-containing catalysts” **Applied Catalysis B: Environmental** 123, 229-240. [10.1016/j.apcatb.2012.04.012](https://doi.org/10.1016/j.apcatb.2012.04.012).

PROFESSIONAL AFFILIATIONS

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| • ASTM International | 2024 — Present |
| • American Chemical Society | 2019 — 2026 |
| • CGSB Petroleum Fuels Board | 2016 — 2025 |
| • Canadian Society for Chemistry | Periodic Member, 2013 — 2025 |